
Software Requirements Specification

for

AutismCare

Version 1.0 approved

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define and describe the complete set of software requirements for the Autism Care Platform – Release 1.0, a web-based MERN stack application designed to facilitate early screening, diagnosis, therapy, laboratory reporting, and developmental monitoring for children under 4 years of age with suspected Autism Spectrum Disorder (ASD).

This document identifies the full product whose requirements are being specified, covering all major subsystems integrated within the Autism Care Platform. It outlines the intended functionality of each module and defines how these modules interact to support the overall workflow of early autism care. The SRS serves as a formal reference for developers, testers, stakeholders, and evaluators to ensure that the system is built according to agreed-upon specifications.

The scope covered by this SRS includes **all primary modules and subsystems** of the Autism Care Platform:

1.1.1 Parent Module

Covers requirements related to child registration, initial ASD screening (questionnaire and AI-based facial feature analysis), appointment booking, receiving diagnosis reports, viewing lab results, accessing therapy plans, and tracking progress over time.

1.1.2 Clinician Module

Includes detailed diagnostic workflows, review of screening results, creation of diagnostic reports, referral to labs/therapists, appointment management, and secure communication with parents and therapists.

1.1.3 Therapist Module

Defines requirements for therapy session management, therapy plan creation, progress updates, session notes, and communication features.

1.1.4 Laboratory Module

Specifies features for receiving test requests, processing samples, uploading lab reports, and forwarding results to clinicians and parents.

1.1.5 Admin Module

Covers administrative tasks such as user role management, approval of clinicians/therapists/lab staff, system configuration, security oversight, and management of screening tools and therapy guidelines.

1.2 Document Conventions

This Software Requirements Specification follows standard IEEE guidelines and adopts specific typographical and semantic conventions to ensure clarity, consistency, and uniform interpretation throughout the document. The following conventions indicate the meaning, priority, and formatting of requirement statements, diagrams, and references used in this SRS.

The keywords “**Shall**,” “**Should**,” and “**May**” are used in accordance with IEEE standards. The term.

- “**Shall**” indicates a **mandatory requirement** that must be fully implemented for the system to be considered compliant. These requirements are binding and non-negotiable.
- “**Should**” denotes a **recommended requirement**, representing desirable system behavior that improves functionality but is not mandatory for the minimum viable release.
- “**May**” indicates an **optional requirement**, which may be implemented depending on feasibility, available resources, or future enhancement plans.
- **Each requirement has its own explicit priority**; priorities of higher-level requirements are **not inherited** by lower-level or detailed requirements unless clearly stated.
- **Bold text and bold section headings** are used to highlight major functional sections and top-level modules within the SRS.
- **Subsections and requirement identifiers** follow a structured hierarchical numbering format (e.g., **1.0, 1.1, 2.3.4**) to ensure clear organization and easy traceability.
- **Use cases** are referenced using the naming convention **UC_#** (e.g., UC_01, UC_02), ensuring each use case is uniquely identifiable.
- **Requirements associated with use cases** reference these UC identifiers to support consistent cross-mapping and traceability across the document.

To maintain uniform document structure, standard English fonts (e.g., Times New Roman or Arial) in consistent sizes are used. Technical terms, variable names, and identifiers may appear in *italic* or *monospace* formatting where needed for clarity.

These conventions ensure that all stakeholders including developers, testers, supervisors, and project evaluators interpret requirements consistently and accurately throughout the development lifecycle.

1.3 Intended Audience and Reading Suggestions

This Software Requirements Specification (SRS) is intended for a diverse group of stakeholders involved in the development, evaluation, and usage of the **Autism Care Platform – Release 1.0**. Each reader may have different interests and responsibilities, and this document provides information relevant to all technical and non-technical roles associated with the project.

The primary intended audiences include:

1.3.1 Developers / Software Engineers

Responsible for designing and implementing the MERN-based system according to the requirements defined in this SRS. They should pay particular attention to the functional requirements, system architecture descriptions, and module-specific behaviors.

1.3.2 Project Managers / Supervisors

Oversee project planning, progress tracking, and resource allocation. They will use this document to understand project scope, constraints, assumptions, and key deliverables.

1.3.3 Quality Assurance Testers:

Use this SRS as the baseline for designing test cases, validation procedures, performance evaluations, and acceptance criteria. Testers should focus on functional requirements, use cases, and non-functional requirements.

1.3.4 Clinicians, Therapists, and Laboratory Personnel (Domain Experts):

These users rely on the system to support screening, diagnosis, therapy management, and reporting. This document helps them understand the workflow automation and capabilities provided by each module.

1.3.5 End Users (Parents/Guardians):

Although this SRS is primarily technical, parents or representatives may refer to the overview and functional sections to understand the intended features affecting their interaction with the system.

1.3.6 Documentation Writers and UI/UX Designers:

This SRS guides the creation of user manuals, interface specifications, help documents, and workflow designs. They should focus on user stories, interface requirements, and module-specific interactions.

1.3.7 Academic Evaluators / FYP Examiners:

Project examiners will use this SRS to assess completeness, correctness, and adherence to IEEE standards.

To support clarity and ease of understanding, readers are advised to follow this recommended reading sequence:

1. **Section 1 – Introduction:** Provides the purpose, scope, definitions, and overall context of the Autism Care Platform. All readers should start here.
2. **Section 2 – Overall Description:** Offers a high-level overview of the system environment, user characteristics, constraints, assumptions, and system landscape. Best for project managers, evaluators, and designers.
3. **Section 3 – System Features / Functional Requirements:** Contains detailed descriptions of each module (Parent, Clinician, Therapist, Lab, Admin) and is essential for developers, testers, and domain experts.
4. **Section 4 – External Interface Requirements:** Describes UI/UX expectations, hardware/software interfaces, and communication protocols. Important for developers and designers.
5. **Section 5 – Non-Functional Requirements:** Provides security, performance, reliability, compliance, and usability requirements. Useful for architects, developers, and QA testers.
6. **Section 6 – System Diagrams:** Includes UML diagrams such as use case diagrams, sequence diagrams, activity diagrams, and class diagrams. Recommended for developers, testers, and academic examiners.

7. Section 7 – Other Requirements / Appendices:

Contains supporting material such as data dictionaries, glossary, references, and AI-processing workflows (e.g., facial analysis pipeline).

This reading order ensures that each stakeholder engages with the sections most relevant to their role while gaining a structured understanding of the Autism Care Platform and its requirements.

1.4 Product Scope

The Autism Care Platform is a comprehensive web-based system developed using the MERN stack with the purpose of supporting early screening, diagnosis, therapy management, laboratory reporting, and continuous developmental tracking for children under four years of age who are suspected of having Autism Spectrum Disorder (ASD). The software centralizes all major components of early autism care into a unified digital platform that connects parents, clinicians, therapists, laboratory staff, and administrators. By integrating AI-based facial feature analysis, standardized screening questionnaires, diagnostic workflows, therapy tracking tools, and progress monitoring dashboards, the system aims to facilitate more accurate evaluations and promote timely intervention for young children.

The primary objective of the platform is to streamline and modernize the autism assessment process by replacing manual, disconnected, and paper-based methods with a centralized, secure, and automated solution. The system is designed to enhance communication among stakeholders, reduce diagnostic delays, improve the organization of clinical and therapy sessions, and provide parents with continuous visibility into their child's developmental progress. In addition, the software seeks to minimize the inconsistencies commonly faced in traditional autism-care environments by offering structured workflows, digital reporting, and data-driven insights.

Beyond its core clinical purpose, the Autism Care Platform aligns with broader academic and technological goals by demonstrating the application of full-stack development, AI-based visual screening, and health-oriented information systems. For the context of a Final Year Project, the platform showcases the integration of modern web technologies, user-centered design, and machine learning techniques within a socially impactful domain. Its intended outcomes include improving accessibility to early autism support, assisting professionals in making evidence-informed decisions, and providing families with an organized and transparent way to manage their child's care. The system may also serve as a foundation for future expansion into mobile applications, advanced analytics, and larger clinical environments.

2. Overall Description

2.1 Product Perspective

The Autism Care Platform is a new, self-contained web-based system developed as a comprehensive and unified solution for the early screening, diagnosis, therapy management, laboratory reporting, and developmental tracking of children under four years of age suspected of having Autism Spectrum Disorder (ASD). Unlike existing solutions that focus on isolated functionalities such as standalone screening tools, therapy scheduling systems, or hospital laboratory portals this platform integrates all major components of early autism care into a single environment. The platform is conceptualized as an independent system but is designed with extensibility in mind, allowing future integration with electronic health record (EHR) systems, national health registries, or mobile health applications. Currently, it serves as a centralized ecosystem connecting parents, clinicians, therapists, and laboratory personnel, streamlining communication, improving diagnostic accuracy, and reducing fragmentation in autism-related healthcare workflows. The product is structured into multiple modules, each representing a distinct subsystem, yet all interacting seamlessly.

through secure APIs and a shared database. These subsystems collectively form a unified web-based solution capable of supporting end-to-end autism management within a digital healthcare framework.

2.1.1 Context Model

The context model for the Autism Care Platform represents how the system interacts with its primary users and external components within the broader organizational and healthcare environment. As a conceptual model, it illustrates the position of the platform within the ecosystem of stakeholders responsible for early autism identification and intervention. The system acts as the central node where parents submit screening data and upload child images, clinicians perform diagnostic evaluations, therapists provide therapy reports, and laboratory staff upload test results. These interactions demonstrate the flow of information between human users and the digital platform, ensuring that every user role contributes to a unified developmental assessment for each child.

In a broader organizational context, the Autism Care Platform aligns with modern healthcare digitization goals by supporting remote assessment, reducing diagnostic delays, and enabling early intervention—all critical factors in autism management. The context model also clarifies the external interfaces, such as potential integration with hospital systems, notification services, or authentication services, although the current version functions as an independent product. The model helps define user boundaries, data flow expectations, and how each subsystem contributes to the platform's overall objective of supporting early ASD detection and continuous developmental monitoring.

2.2 Product Functions

The Autism Care Management System provides an integrated set of functions that enable parents, clinicians, therapists, laboratory staff, and administrators to collaboratively support early screening, diagnosis, therapy, and monitoring of children with suspected Autism Spectrum Disorder (ASD). The system allows parents to register their child, submit assessments, request appointments, receive reports, access educational material, and receive system-generated notifications. Clinicians use the platform to manage child profiles, prescribe diagnostic tests, review laboratory results, schedule and manage appointments, and monitor the child's developmental progress over time. Therapists interact with the system to record session logs, upload home-based tasks, update therapy progress, and communicate with parents and clinicians. Laboratory technicians use the platform to upload test reports, share results with clinicians, and receive diagnostic test prescriptions. The admin module manages users, roles, system access, and generates analytics reports to ensure smooth workflow and compliance. All modules integrate tightly with the central database, ensuring secure, real-time data exchange across all user types.

Feature 1: Parent Registration and Child Profile Management

Enables parents to create accounts, register child profiles, and manage personal and developmental information.

Feature 2: Initial ASD Screening (Questionnaire + AI-Based Facial Analysis)

Allows parents to perform early autism screening using standardized questionnaires and facial morphology analysis tools.

Feature 3: Appointment Booking and Scheduling

Lets parents request appointments with clinicians and therapists; clinicians can approve, reject, or reschedule appointments.

Feature 4: Clinician Diagnostic Assessment

Clinicians can review screening results, conduct detailed assessments, generate diagnostic reports, and prescribe therapy or lab tests.

Feature 5: Laboratory Test Requests and Report Uploads

Clinicians can send lab test requests, and lab technicians can upload test reports, which become accessible to parents and clinicians.

Feature 6: Therapy Session Management

Therapists can manage therapy schedules, upload session logs, assign home tasks, and update the child's progress.

Feature 7: Progress Tracking and Report Viewing

Parents can view diagnostic results, therapy outcomes, lab reports, and graphical progress charts of the child's development.

Feature 8: Notifications and Educational Materials

The system sends reminders, updates, and educational resources to parents regarding ASD awareness and child care.

Feature 9: Admin User Management and Verification

Admin verifies and approves clinician, therapist, and lab technician accounts, manages user roles, and maintains system integrity.

Feature 10: System Analytics and Monitoring

Admin can view system activity logs, usage reports, appointment statistics, and performance metrics.

Feature 11: Secure Database Management

Stores all user profiles, assessments, reports, session logs, and communication records securely in the centralized database.

2.2.1 Data flow diagram (Functional Model)**Context Level DFD**

The context-level Data Flow Diagram represents the Autism Care Management System as a single, unified process interacting with five external entities: Parent, Clinician, Therapist, Laboratory, and Admin. Each external entity exchanges specific information with the system based on their operational role.

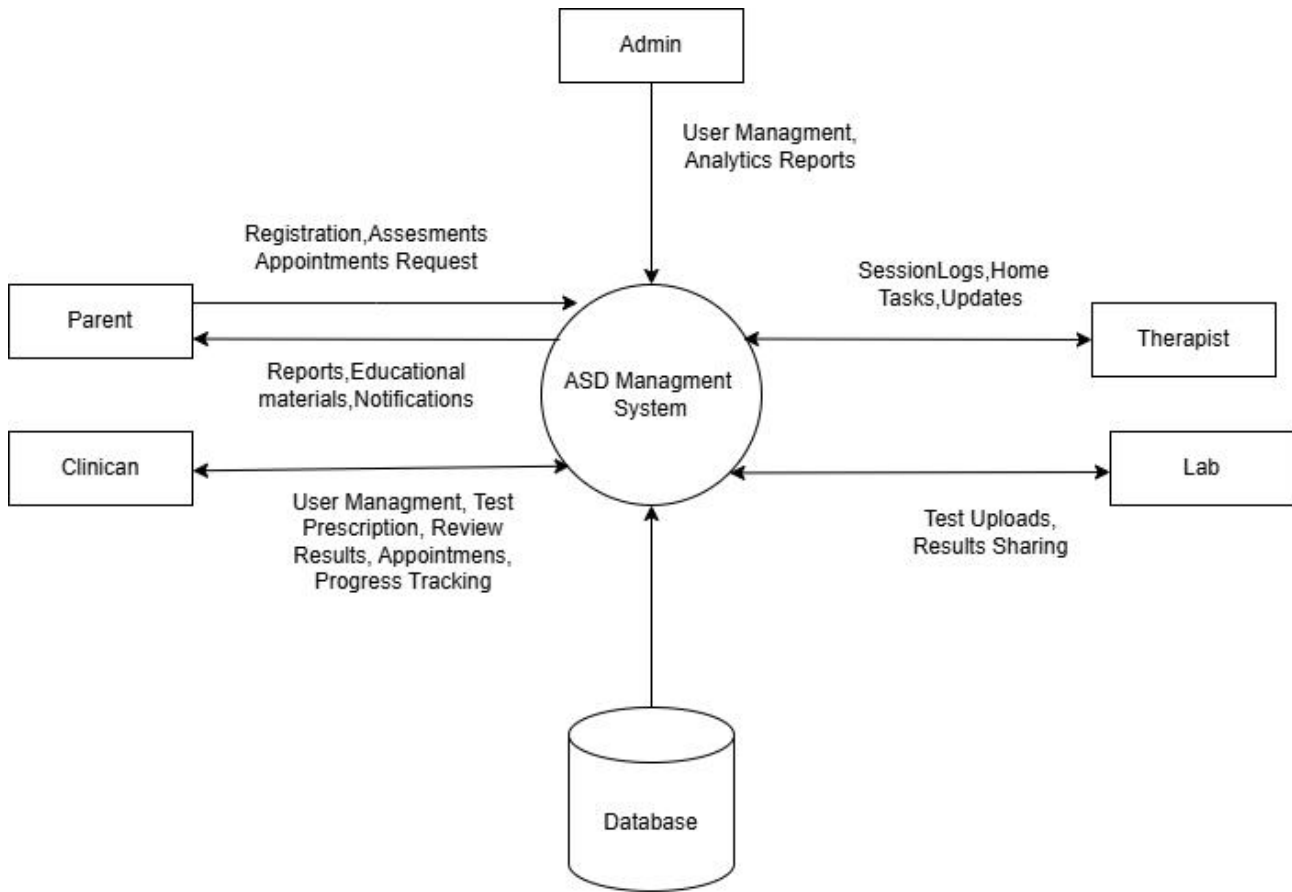
Parents interact with the system by submitting child registrations, completing assessments, and requesting appointments. In return, they receive diagnostic reports, educational materials, and automated notifications. Clinicians contribute to the system by managing child cases, prescribing diagnostic tests, reviewing laboratory results, overseeing appointments, and tracking the child's progress. The system, in return, provides clinicians with updated test results, therapy feedback, and parent-submitted assessments.

Therapists send session logs, therapy updates, and home-based tasks to the system, while receiving clinician recommendations and child information needed to personalize therapy plans. Laboratories receive test prescriptions from clinicians through the system and return completed test uploads and results, which are automatically routed to both clinicians and parents.

The admin user performs high-level management tasks such as user management, staff approvals, and system analytics reporting. The system responds by providing activity logs, analytics summaries, and access-control features. All data generated through these interactions is stored, retrieved, and processed through a centralized database to maintain integrity and real-time access across the platform.

This context model shows the system's functional boundary and clearly describes how external actors contribute and retrieve information from the Autism Care Management System in a unified, streamlined workflow.

Figure 1: Context Level Data Flow Diagram for Autism Care Platform



2.3 User Classes and Characteristics

The “Autism Care” platform serves multiple categories of users, each with distinct functional needs, technical expertise, access privileges, and frequency of interaction. Proper identification of user classes and external entities is essential for designing a secure, efficient, and user-centered system.

The system distinguishes the following user classes:

1. **Parents / Guardians**

Parents are the primary users of the platform. They typically have low to medium technical expertise and interact with the system frequently. Their main functions include: registering their child, completing initial screening questionnaires, uploading facial images for AI-based assessment, booking appointments with clinicians or therapists, and tracking their child’s progress through reports and dashboards. Parents’ access is limited to their child’s data, and they do not engage in administrative or clinical decision-making. Ensuring usability and privacy for this class is critical.

2. **Clinicians (Psychologists / Developmental Specialists)**

Clinicians are highly trained professionals responsible for reviewing screening results, performing detailed diagnostics, and generating formal assessments. They require access to confidential patient data, lab reports, historical progress, and AI-based analysis outcomes. Their technical expertise is medium to high, and they interact with the system regularly. Clinicians are among the most important user classes, as accurate diagnosis forms the core purpose of the system.

3. **Therapists (Speech, Occupational, Behavioral)**

Therapists operate after clinicians provide a diagnosis. They update therapy session notes, record progress metrics, and provide feedback to parents. Their technical expertise is medium, and they interact

with the system on a scheduled basis. Therapists have access restricted to children assigned to them, and they do not perform screening or administrative tasks.

4. **Lab Technicians**

Lab technicians interact with the system occasionally, primarily for uploading diagnostic test results requested by clinicians. They have limited technical skills and access privileges restricted to lab-specific tasks. They do not access therapy, screening, or administrative modules.

5. **Administrators**

Administrators manage system-wide operations, including user role approvals, security monitoring, configuration management, and AI dataset oversight. They are technically proficient and have high-level privileges. Administrators do not directly engage in screening, therapy, or diagnostics, but their role ensures operational integrity and compliance with privacy standards.

6. **Children (Data Subjects)**

Children do not interact directly with the system but are central to the platform as the data subject. All information—screening responses, facial images, diagnostic results, therapy sessions, and progress reports—is linked to them. Their data requires strict privacy, controlled access, and adherence to ethical and legal standards.

7. **Importance Hierarchy:** Parents, clinicians, and therapists are the primary user classes. Lab technicians and administrators are essential for support and security, while children are indirect but critical data subjects.

2.3.1 Identifying External Entities

External entities represent actors or systems that interact with the platform but are outside its boundary. The identification is performed in two phases:

2.3.1.1 Phase 1: Over-Specification of Entities

Based on the abstract and system modules, potential external entities include:

- Parent / Guardian
- Child
- Clinician
- Therapist
- Lab Technician
- Administrator
- Screening Questionnaire
- Facial Image / AI Module
- Appointment System
- Therapy Session Records
- Progress Report / Tracker
- Lab Report
- Notifications System

This phase intentionally includes all conceivable entities, ensuring nothing is missed before refinement.

2.3.1.2 Phase 2: Refinement Based on Business Logic

After considering the actual workflow and business rules, the list is refined to include only entities directly relevant to the platform's operations:

- **Parent** – initiates screening, uploads data, and monitors progress
- **Child** – central data subject
- **Clinician** – performs diagnosis and approves reports
- **Therapist** – updates therapy progress
- **Lab Technician** – uploads lab reports

- **Administrator** – manages user roles and system operations
- **AI Screening Engine** – processes facial images and questionnaire responses
- **Appointment Manager** – schedules and tracks sessions
- **Notification System** – sends alerts, reminders, and updates

These refined entities align directly with the core functional requirements and maintain system simplicity, security, and efficiency.

2.4 Operating Environment

The Autism Care platform is a **web-based system built on the MERN (MongoDB, Express.js, React.js, Node.js) stack**, designed to provide seamless accessibility, high performance, and secure management of sensitive data related to children under four years of age. The system operates in an environment that accommodates multiple user classes, including parents, clinicians, therapists, lab technicians, and administrators, each with varying access privileges and usage patterns.

1. Hardware Environment:

- Users can access the system via desktop computers, laptops, tablets, and smartphones with standard input/output devices (keyboard, mouse, touch screen).
- The server environment requires a reliable cloud-based or on-premises server with sufficient CPU, memory, and storage to handle concurrent users, AI-based facial image processing, database operations, and file uploads.
- Lab technicians may use imaging devices or scanners to digitize reports before uploading to the system.

2. Software Environment:

- **Client-side:** Modern web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari are supported. The user interface is responsive, enabling accessibility on both desktop and mobile devices.
- **Server-side:** Node.js runtime with Express.js framework handles backend logic and API requests.
- **Database:** MongoDB is used for storing structured and unstructured data including user profiles, screening results, therapy logs, lab reports, and progress tracking.
- **AI/ML Modules:** Facial image analysis is performed using integrated AI modules (e.g., Convolutional Neural Networks) on the server side or via cloud-based AI services.
- **Operating Systems:** Server can run on Linux (Ubuntu, CentOS) or Windows Server, while client devices can use Windows, macOS, iOS, Android, or Linux OS.

3. Network Environment:

- The system is designed for internet connectivity over TCP/IP networks.
- HTTPS protocol ensures secure data transmission between clients and the server.
- Optional internal VPN access may be provided for administrators or clinicians working in restricted environments.

4. Constraints and Dependencies:

- High-speed internet is recommended for smooth uploading of facial images and downloading of reports.
- The system relies on third-party email/SMS services for notifications and reminders.
- Compliance with data privacy laws (e.g., GDPR, HIPAA, or local regulations) is mandatory due to sensitive child health data.

2.5 Design and Implementation Constraints

The design and implementation of the Autism Care platform are subject to a number of constraints that affect the choices of developers and the overall system architecture. These constraints are driven by regulatory requirements, technical limitations, and operational considerations specific to the healthcare and child development domain.

2.5.1 Regulatory and Privacy Constraints:

- The system handles sensitive personal data of children under four years of age, including facial images, screening responses, diagnostic results, and therapy progress. Compliance with local data protection laws, GDPR, and HIPAA is mandatory.
- Parental consent must be obtained prior to data collection, storage, or processing through AI modules.
- Secure handling of data, including encryption, access control, and audit trails, is required to prevent unauthorized access.

2.5.2 Hardware and Performance Constraints:

- The platform must support multiple concurrent users (parents, clinicians, therapists, lab technicians, and administrators) without performance degradation.
- AI-based facial image analysis may require GPU acceleration or high-performance computing resources for timely processing.
- Sufficient storage must be provisioned for high-resolution facial images, historical reports, therapy records, and lab reports.

2.5.3 Software and Technology Constraints:

- The platform is built using the MERN stack (MongoDB, Express.js, React.js, Node.js), which requires developers to work primarily with JavaScript/TypeScript.
- AI and machine learning modules for facial feature analysis are implemented in Python frameworks such as TensorFlow or PyTorch. Communication between backend and AI modules must be handled through APIs.
- The application must support modern web browsers (Chrome, Firefox, Edge, Safari) and be responsive for desktop, tablet, and mobile devices.

2.5.4 Interface and Integration Constraints:

- The system integrates with third-party services such as email/SMS notification providers and may optionally use cloud-based AI services, requiring adherence to standardized API protocols (REST/HTTPS).
- Lab reports may come in multiple formats (PDF, images), requiring standardized parsing and database storage.
- Integration with appointment scheduling and therapy modules must follow consistent API and data format standards.

2.5.5 Security Constraints:

- Multi-level access control must ensure that users can access only the data and modules relevant to their roles.
- All communication between clients and the server must use secure protocols (HTTPS).
- Sensitive information, including passwords and personally identifiable data, must be encrypted using industry-standard algorithms such as AES or bcrypt.

2.5.6 Design and Programming Standards:

- The system follows the MVC (Model-View-Controller) architecture to separate presentation, logic, and data layers.
- Developers must ensure code modularity, readability, and maintainability, as the customer organization may be responsible for future updates.
- Database design must support scalability, efficient queries, and integration with AI analytics and reporting modules.

2.5.7 Operational Constraints:

- Some functionalities, including AI-based screening and appointment scheduling, require a stable internet connection and cannot operate offline.
- The system must perform reliably under variable network conditions and on devices with limited processing power.

2.6 User Documentation

To ensure effective use of the Autism Care platform, comprehensive user documentation will be provided for all user classes, including parents, clinicians, therapists, lab technicians, and administrators. The documentation will guide users through system functionalities, features, and best practices while supporting compliance with privacy and security standards.

2.6.1 User Manuals:

- **Parent Manual:** Provides step-by-step instructions on registration, creating child profiles, completing screening questionnaires, uploading facial images, booking appointments, and viewing progress reports. Emphasizes privacy and consent management.
- **Clinician Manual:** Covers diagnostic workflow, reviewing AI-generated screening results, conducting detailed assessments, generating reports, and tracking patient history. Includes guidelines on interpreting AI outputs and lab report integration.
- **Therapist Manual:** Explains therapy session management, updating progress, communication with parents, and documentation of therapy notes.
- **Lab Technician Manual:** Details the process for receiving test requests, uploading lab reports, and updating report status.
- **Administrator Manual:** Provides guidance on user management, role assignment, system monitoring, security oversight, and AI model management.

2.6.2 Online Help and Tutorials:

- Context-sensitive online help will be integrated within the platform to provide instant guidance for tasks such as screening, booking appointments, uploading files, and generating reports.
- Interactive tutorials for parents and professionals will demonstrate key workflows, including AI-based screening, appointment scheduling, and therapy tracking.

2.6.3 Delivery Formats:

- Documentation will be provided in **PDF format** for offline access.
- Online help and tutorials will be **web-integrated**, accessible through the platform on desktop and mobile devices.
- Video tutorials will be provided for complex workflows, such as facial image uploads, AI screening interpretation, and therapy progress tracking.

2.6.4 Standards and Guidelines:

- All documentation will follow clear, concise, and user-friendly writing standards.
- Visual aids such as screenshots, flow diagrams, and step-by-step illustrations will be included to enhance understanding.
- Documentation will comply with accessibility guidelines to support users with diverse technical skills and abilities.

2.7 Assumptions and Dependencies

The development and successful operation of the Autism Care platform rely on several assumptions and dependencies that may influence the requirements and system performance. These factors include technology availability, user behavior, regulatory compliance, and integration with third-party services.

2.7.1 Assumptions

2.7.1.1 User Access and Competence:

Parents, clinicians, therapists, and lab technicians have basic proficiency in using web-based applications and can access the system using standard devices such as desktops, laptops, tablets, or smartphones. Users will provide accurate and complete information during registration and screening processes.

2.7.1.2 Parental Consent:

All data collected from children under four years of age will be provided with explicit parental or guardian consent. Users will comply with privacy requirements and terms of service for data submission.

2.7.1.3 Internet Connectivity:

The system assumes stable internet access for core functionalities, including AI-based facial image processing, appointment scheduling, report uploads, and notifications. Low bandwidth or offline access may limit some features but will not affect the integrity of stored data.

2.7.1.4 Hardware and Device Requirements:

Client devices have minimum processing power, memory, and display resolution to interact with the responsive web interface and view AI-generated screening results and reports. Server infrastructure will meet the necessary performance requirements to process AI models, store large datasets, and support concurrent users.

2.7.1.5 Regulatory Compliance:

The system assumes continued adherence to data protection regulations (e.g., GDPR, HIPAA, or applicable local laws) during both development and operation phases.

2.7.2 Dependencies

2.7.2.1 Third-Party Components:

The platform depends on third-party services for AI-based facial image analysis, which may be hosted either on the cloud or integrated locally. Integration with email and SMS notification services relies on external providers using standardized APIs.

2.7.2.2 Software Stack Dependencies:

The MERN stack (MongoDB, Express.js, React.js, Node.js) is assumed to be stable, compatible with deployed environments, and actively maintained. Python-based AI/ML frameworks (e.g., TensorFlow or PyTorch) are required for facial feature analysis and must remain compatible with backend services.

2.7.2.3 Data Availability:

The system depends on parents providing high-quality images and completed screening questionnaires for AI analysis. Lab reports and clinical data must follow standard formats to ensure correct storage and integration.

2.7.2.4 Maintenance and Updates:

The system's functionality depends on timely updates of libraries, frameworks, and server software to maintain security, performance, and compatibility. Any changes in AI models, third-party services, or regulatory requirements may require adjustments to system features.

3. External Interface Requirements

The Autism Care platform provides a web-based user interface designed for various user classes, including parents, clinicians, therapists, lab technicians, and administrators. The interface aims for intuitive navigation, accessibility, and responsive design across multiple device types.

3.1 User Interfaces

The Autism Care platform provides a web-based user interface designed for various user classes, including parents, clinicians, therapists, lab technicians, and administrators. The interface aims for intuitive navigation, accessibility, and responsive design across multiple device types.

3.1.1 Logical Characteristics:

- Consistent layout and navigation across all modules.
- Responsive design for desktops, tablets, and smartphones.
- Standard UI elements such as buttons, dropdown menus, checkboxes, radio buttons, text fields, and tooltips.
- Context-sensitive Help buttons on all screens.
- Standard error messages with actionable guidance (e.g., “Please enter a valid email address”).
- Keyboard shortcuts for frequently used actions (e.g., Ctrl+S to save forms).
- Visual indicators for required fields, warnings, and success messages.
- Accessibility features: screen reader compatibility, high-contrast mode, and tab navigation.

3.1.2 Software Components with UI:

- **Child Profile & Screening Module** – for data entry, image upload, and questionnaire completion.
- **AI Screening Module** – review of AI-generated results by clinicians; summary view for parents.
- **Appointment Module** – scheduling, updating, or canceling appointments by parents, clinicians, and therapists.
- **Therapy Module** – session management and progress tracking by therapists; progress viewing by parents.
- **Lab Report Module** – upload by lab technicians; review by clinicians.
- **Administration Module** – user and role management, system monitoring by administrators.
- **Notification Module** – alerts, reminders, and messages for all users.

3.2 Hardware Interfaces

The Autism Care system interacts with various hardware components to facilitate user operations, data processing, and storage.

3.2.1 Supported Device Types:

- Client devices: desktops, laptops, tablets, smartphones.
- Server: cloud-hosted or on-premises servers with sufficient CPU, GPU (for AI processing), memory, and storage.
- Lab equipment: scanners or imaging devices for digitizing lab reports.

3.2.2 Data and Control Interactions:

- Image uploads from parent devices to the server for AI-based analysis.
- Appointment requests, report uploads, and therapy session data transmitted between client devices and server.
- Server responses delivered to client interfaces in real-time.

3.2.3 Communication Protocols:

- HTTPS for secure web communication.
- REST APIs for communication between backend services, AI modules, and front-end interfaces.
- TCP/IP for standard network connectivity.

3.3 Software Interfaces

The Autism Care system integrates with several software components to provide full functionality.

3.3.1 Primary Software Components:

- **MongoDB (Database)** – stores user data, child profiles, screening results, lab reports, therapy records, and AI analysis outputs.
- **Node.js with Express.js (Backend)** – handles API requests, business logic, and integration with AI modules.
- **React.js (Frontend)** – renders dynamic web interfaces for all user roles.
- **Python AI/ML Frameworks (TensorFlow/PyTorch)** – process facial images and questionnaire data for screening analysis.
- **Third-Party Services:** Email/SMS notification providers; optional cloud-based AI services.

3.3.2 Data Items and Messages:

- Incoming: child profiles, screening questionnaires, facial images, appointment requests, lab report uploads.
- Outgoing: AI screening results, appointment confirmations, therapy progress summaries, notifications, and alerts.

3.3.3 Data Sharing and Implementation Constraints:

- RESTful APIs connect Node.js backend with Python AI modules.
- JSON format is used for data exchange between frontend, backend, and AI services.
- Shared data stored in MongoDB with strict role-based access control.

3.4 Communications Interfaces

The system relies on secure and reliable communication channels to facilitate interactions between users and the backend services.

3.4.1 Requirements:

- **Web Communication:** HTTPS protocol for all client-server interactions.
- **Notifications:** Email and SMS for reminders, alerts, and confirmations via standardized APIs.
- **Data Transfer:** High-resolution images and lab reports transmitted securely, with encryption in transit.
- **Message Formatting:** JSON for API payloads; PDF/image formats for lab reports.
- **Security and Encryption:** SSL/TLS encryption for all communications; role-based access ensures secure data exchange
- **Data Transfer Rates and Synchronization:** Optimized for smooth uploads and downloads; asynchronous processing for AI analysis and lab report handling.

4. System Features

<This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product. You may prefer to organize this section by use case, mode of operation, user class, object class, functional hierarchy, or combinations of these, whatever makes the most logical sense for your product.>

4.1 SYSTEM FEATURES & INITIAL REQUIREMENTS

4.1.1 Feature 1 :Parent Registration & Authentication

Table 1.0:Parent Registration & Authentication

Para #	Initial Requirement	Description
1.0	The system shall allow parents to create an account.	Parents must register by providing verified information (CNIC, child details, email, phone).
1.1	The system shall support secure login for parents.	Login via email/password; password hashing required.
1.2	The system shall provide password recovery.	Reset via email OTP.
1.3	The system shall verify parent identity before enabling screening and booking.	System checks CNIC/phone verification.

4.1.2 Feature 2 : Initial Screening (Questionnaire + AI Facial Analysis)

Table 1.1:Initial Screening (Questionnaire + AI Facial Analysis)

Para #	Initial Requirement	Description
2.0	The system shall provide autism screening questionnaire for children <4 years.	Includes behavior, speech, and developmental questions.
2.1	The system shall allow parents to upload the child's image for AI-based facial feature extraction.	Used for autistic morphology detection.
2.2	The AI module shall analyze facial features to detect autism-related markers.	CNN/ML model identifies facial patterns.
2.3	The system shall generate a combined screening score (questionnaire + AI analysis).	Output: "Low / Medium / High Risk."
2.4	The system shall store screening history for future clinical diagnosis.	Stored securely and encrypted.

4.1.3 Feature 3 :Appointment Booking Module

Table 1.2:Appointment Booking Module

Para #	Initial Requirement	Description
3.0	The system shall allow parents to book appointments with clinicians.	Date, time, clinician selection.
3.1	The system shall show clinician availability slots.	Dynamic schedule.
3.2	Parents shall be able to cancel or update appointments before clinician confirmation.	Allowed until clinician approves.
3.3	The system shall send booking notifications.	SMS/Email/Portal alert.

4.1.4 Feature 4: Clinician Dashboard & Diagnosis Module

Table 1.3:Clinician Dashboard & Diagnosis Module

Para #	Initial Requirement	Description
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4.0	Clinician shall view pending appointments.	Shows time, parent info, screening results.
4.1	Clinician shall perform a detailed clinical diagnosis.	Behavioral checklists, developmental milestones.
4.2	Clinician shall upload clinical diagnosis reports.	PDF or direct entry form.
4.3	Clinician shall assign therapy plan if needed.	Speech/occupational therapy scheduling.

4.1.5 Feature 5 :Therapist Module

Table 1.4:Therapist Module

Para #	Initial Requirement	Description
5.0	Therapist shall access assigned patients.	Based on clinician referral.
5.1	Therapist shall track therapy progress.	Session notes, improvement areas.
5.2	Therapist shall update progress reports after each session.	Stored for parent view.

4.1.6 Feature 6 :Laboratory Module

Table 1.5:Laboratory Module

Para #	Initial Requirement	Description
6.0	Lab technician shall access clinician test requests.	EEG, genetic, behavioral tests.
6.1	Lab technician shall upload lab reports.	PDF or structured format.
6.2	The system shall notify parents and clinicians upon report upload.	Real-time alerts.

4.1.7 Feature 7: Admin Module (Approval + System Control)

Table 1.6:Admin Module (Approval + System Control)

Para #	Initial Requirement	Description
7.0	Admin shall approve/reject clinician/therapist registrations.	Based on licenses and credentials.
7.1	Admin shall manage all users and system roles.	CRUD operations.
7.2	Admin shall monitor system performance logs.	Security, usage.
7.3	Admin shall take action on complaints or misuse.	User moderation.

4.2 Allocate Requirements

Table 2: Allocate Requirements

Para #	Initial Requirement	Use Case Name
1.0	Parent registration	UC Parent Registration
1.1	Parent login & authentication	UC Parent Login
2.0	Screening questionnaire	UC Screening Questionnaire
2.1	Facial feature upload	UC Upload Facial Image
2.2	AI analysis	UC AI Facial Analysis
3.0	Book appointment	UC Book Appointment
3.2	Cancel/update appointment	UC Manage Appointment

4.0	Clinician handles appointment	UC Clinician View Appointment
4.1	Clinician diagnosis	UC Clinical Diagnosis
4.3	Assign therapy plan	UC Assign Therapy
5.1	Therapist update progress	UC Therapy Progress Update
6.1	Lab technician uploads report	UC Lab Report Upload
7.0	Admin approves clinician/therapist	UC Admin Approval

4.3 Prioritize Requirements

Table 3: Prioritize Requirements

Para #	Rank	Requirement	Use Case ID	Use Case Name
1.0	Highest	Parent Registration	UC_1	UC_Parent_Registration
1.1	Highest	Parent Login	UC_2	UC_Parent_Login
2.0	Highest	Screening Questionnaire	UC_3	UC_Screening_Questionnaire
2.1	High	Facial Image Upload	UC_4	UC_Upload_Facial_Image
2.2	High	AI Screening	UC_5	UC_AI_Facial_Analysis
3.0	Highest	Appointment Booking	UC_6	UC_Book_Appointment
4.1	Highest	Clinical Diagnosis	UC_7	UC_Clinical_Diagnosis
5.1	Medium	Therapy Progress Update	UC_8	UC_Therapy_Progress_Update
6.1	High	Lab Report Upload	UC_9	UC_Lab_Report_Upload
7.0	Highest	Admin Approval	UC_10	UC_Admin_Approval

4.4 Requirements Traceability Matrix

Table 4: Requirement Traceability Matrix

Sr. No.	Para #	System Requirement	Build	Use Case	Category
1	1.0	Parent Registration	B1	UC_Parent_Registration	Business
2	1.1	Parent Login	B1	UC_Parent_Login	Business
3	2.0	Screening Questionnaire	B2	UC_Screening_Questionnaire	Functional
4	2.1	Facial Image Upload	B2	UC_Upload_Facial_Image	AI Module
5	2.2	AI Screening	B3	UC_AI_Facial_Analysis	AI Module
6	3.0	Appointment Booking	B3	UC_Book_Appointment	Functional
7	4.1	Clinical Diagnosis	B4	UC_Clinical_Diagnosis	Clinical
8	5.1	Therapy	B4	UC_Therapy_Progress_Update	Therapy

		Progress			
9	6.1	Lab Report Upload	B4	UC_Lab_Report_Upload	Laboratory

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Table 5.0:Performance Requirement

ID	Requirement	Rationale / Notes	Acceptance Test	Criteria
PERF-01	System shall authenticate a user (login) in ≤ 2 seconds under normal load (≤ 500 concurrent users).	Fast login improves UX and reduces abandonment.	Measured median latency $\leq 2s$ over 1 hour test with 500 concurrent virtual users.	
PERF-02	Dashboard (Parent/Clinician) shall load initial summary (first paint) in ≤ 3 seconds.	Parents expect quick access to reports.	95th percentile response time $\leq 3s$ under 1k concurrent users.	
PERF-03	AI facial image preprocessing + inference shall complete in ≤ 10 seconds per image (standard model).	Timely screening results are needed for UX / clinician triage.	End-to-end measured $\leq 10s$ for 95% of sample images.	
PERF-04	Upload of lab report ($\leq 25MB$) shall complete with server acknowledgement in ≤ 8 seconds on 10 Mbps connection.	User expectation for file upload responsiveness.	Upload time measured under test network $\leq 8s$.	
PERF-05	Notifications for critical events (high-risk screening) shall be dispatched to clinician within ≤ 60 seconds.	Timely clinician notification is required for risk mitigation.	End-to-end notification latency $\leq 60s$ in 95% tests.	
PERF-06	System shall support scalability to 10,000 registered users with autoscaling strategy.	Projected growth & adoption.	Load test shows linear scaling with additional instances; no degradation beyond 10% in response times as load doubles.	

5.2 Safety Requirements

Table 5.1:Safety Requirements

ID	Requirement	Rationale / Implementation Notes	Verification
SAFE-01	The system shall prevent any automated action that leads to unauthorized medical treatment scheduling without clinician approval.	Avoids harm from automated false positives. Policy: AI suggestions are advisory only.	Review of workflows + UI confirms clinician sign-off required before therapy/test orders.
SAFE-02	The system shall display a clear disclaimer when screening results are not a medical diagnosis and advise clinical follow-up.	Avoids misinterpretation by parents.	UI inspection & user acceptance tests (UAT).
SAFE-03	Emergency contact action: if clinician	Protects patient when	Simulated clinician

	flags emergency, system shall present parent with emergency instructions and contact numbers.	urgent follow-up needed.	emergency flow tested end-to-end.
SAFE-04	The system shall prevent image processing for children unless explicit parental consent is recorded.	Legal & ethical safety.	Consent record presence check before image processing in code review and tests.
SAFE-05	Model failures or low-confidence outputs shall be labeled “Inconclusive” and routed for clinician review.	Prevents misleading automated guidance.	Test cases with low-confidence inputs produce “Inconclusive” and queue clinician task.

Refer to local regulations (e.g., medical device rules, data protection laws) before deployment—include certification planning if the AI is intended for diagnostic claims.

5.3 Security Requirements

Table 5.2: Security Requirement

ID	Requirement	Details / Controls	Test / Evidence
SEC-01	Data in transit shall be protected by TLS 1.2+ (HTTPS).	All endpoints require HTTPS; HSTS recommended.	Security scan / penetration test.
SEC-02	Sensitive personal data and images shall be encrypted at rest (AES-256).	Use KMS for key management; key rotation policy.	DB & storage config review; key rotation logs.
SEC-03	Role-based access control (RBAC) implemented for all operations.	Minimum privilege principle; clinician/admin/therapist/parent roles defined.	RBAC unit tests; access control matrix traceability.
SEC-04	Multi-factor authentication (MFA) required for clinician and admin accounts.	Adds account protection for privileged users.	MFA enabled in configuration and tested.
SEC-05	Audit logging for create/update/delete of PHI records with userID, timestamp, IP.	For forensic and compliance purposes.	Audit log review for sample operations.
SEC-06	Session timeout: auto-logout after 15 minutes of inactivity for clinicians and therapists.	Reduces risk of unauthorized access.	Session management tests.
SEC-07	Vulnerability management: monthly vulnerability scans and quarterly penetration tests.	Ongoing security posture.	Scan & pentest reports.
SEC-08	Data minimization: only collect required fields; privacy-by-design.	Limit exposure of PHI.	Data schema review.
SEC-09	Consent management with auditable records (who consented, when, scope).	Legal compliance.	Consent database records and UI audit.
SEC-10	Backup & disaster recovery: daily backups, tested restore within 24 hours RTO.	Ensures availability and data integrity.	Backup logs + restore test evidence.

5.4 Software Quality Attributes

Table 5.3: Software Quality Attributes

Attribute	Requirement (quantified where possible)	Rationale
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Availability	99.5% uptime (excluding scheduled maintenance).	Clinical workflows rely on availability.
Reliability	Mean time between failures (MTBF) target > 30 days; critical transaction retry logic in place.	Ensures dependable operation.
Maintainability	Codebase shall have unit test coverage $\geq 70\%$ (business logic), CI/CD pipelines for automated tests.	Easier future development and bug fixes.
Performance	See Section 5.1.	Performance expectations.
Usability	Parent tasks (register + start screening) should be completable within 15 minutes by non-technical user; critical flows use stepper UI.	User acceptance.
Portability	Web frontend should be compatible with modern browsers (last 2 versions) and responsive on mobile browsers.	Widest user reach.
Interoperability	System shall support export/import of reports in PDF and HL7 FHIR (basic) format for clinician systems.	Integration with external systems.
Security	See Section 5.3.	Protect PHI.
Testability	All major features (screening, AI inference, upload) shall have automated tests and fixtures.	Ensures regressions are caught.
Explainability	AI module shall produce inference metadata (model version, feature importances when feasible) for clinician review.	Clinical trust & audit.

5.5 Business Rules

Table 5.4: Business Rules

BR-ID	Business Rule	Trigger / Enforcement
BR-01	Clinician must approve therapy plan before therapist executes it.	Enforced by RBAC and workflow state transitions.
BR-02	Child must be < 4 years for automated screening recommendations; if ≥ 4 , system allows but marks as “out-of-scope.”	UI validation + warning prompt.
BR-03	Clinician credentials (license) must be validated by admin before full system access.	Admin approval workflow.
BR-04	Parents can cancel appointments up to 24 hours before without penalty.	Appointment cancellation rules enforced in scheduler.
BR-05	AI results are advisory; only clinicians issue formal diagnoses.	Policy statement displayed in UI and enforced through role permissions.
BR-06	High-risk screening triggers immediate clinician notification and places patient into clinician review queue.	System automated notifications.
BR-07	Lab reports are visible to parent only after clinician releases them OR immediate by default (configurable per deployment).	System config setting unlocks visibility.
BR-08	Privileged (e.g., research) accounts must obtain separate consent for data reuse.	Consent management flow and data export controls.

6. Other Requirements

6.1 Database Requirements

The Autism Care system will operate as a Progressive Web Application built on the MERN stack, with MongoDB serving as the primary database for structured and semi-structured data, including user accounts, appointments, screening results, child records, clinical notes, and therapy logs. All documents will use UUIDs as globally unique identifiers, while Mongoose will enforce schema validation, referential consistency, and integrity rules. Each record will include soft-delete flags and automatic timestamps to maintain accurate audit trails. Metadata for uploaded images, laboratory reports, and PDFs will be stored in MongoDB, while the actual files will reside in an S3-compatible object storage service. To support offline functionality, the PWA will leverage IndexedDB to cache user sessions, recently accessed child records, and queued offline submissions, with sensitive data encrypted until synchronization with the server. The system will implement daily incremental backups and full weekly backups with a default retention period of five years for pediatric health information. MongoDB indexing on high-frequency query fields such as child ID, clinician ID, and appointment datetime will ensure efficient query performance as the platform scales.

6.2 Internationalization (i18n) and Localization (l10n)

The platform will be implemented in English, with all user interface text stored in resource files to enable consistent and maintainable content. All timestamps will be stored in UTC and displayed according to the user's local timezone, with Asia/Karachi set as the default for deployments in Pakistan. The system will follow standard conventions for names, numerals, and dates in English. Translation caching and offline considerations for a PWA will still be applied to ensure that the English content remains accessible even when the user is offline.

6.3 Reuse Objectives

The architecture of the Autism Care platform will emphasize code and component reusability. The AI inference module, responsible for facial feature analysis and preliminary autism screening, will be implemented as an independent microservice so it can be reused across different pediatric or clinical systems in the future. Similarly, the platform's RESTful API design will ensure that the backend services can be integrated into future mobile applications, extended for research platforms, or reused across external healthcare systems without major redesign.

6.4 Other Constraints

The system will primarily be developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), aligning with the original design direction of the Final Year Project. While PostgreSQL may be used for structured transactions, MongoDB may remain an optional component for flexible record storage if required by specific functional needs such as storing variable-length therapy notes or emotional behavior logs. The system's frontend must support modern browsers such as Google Chrome, Firefox, and Safari on both desktop and mobile devices to ensure accessibility for parents and clinicians. Deployment will follow a cloud-first strategy using virtualization and containerization technologies. Docker containers will encapsulate backend and AI services, while Kubernetes or a similar orchestration tool will be used to achieve scalability, resilience, and load balancing across services.

Appendix A: Glossary

AI : *Artificial Intelligence*: A field of computer science that enables machines to simulate human intelligence, such as learning, reasoning, and problem-solving. In this system, AI is used for facial feature analysis and preliminary autism screening.

API : *Application Programming Interface*: A set of protocols and tools that allow different software components to communicate. The Autism Care platform exposes RESTful APIs for backend services.

CRUD : *Create, Read, Update, Delete*: The four basic operations for managing data in a database.

DBMS : *Database Management System*: Software used to create, manage, and query databases. In this system, MongoDB serves as the primary DBMS.

FHIR : *Fast Healthcare Interoperability Resources*: A standard describing data formats and elements for exchanging electronic health records. Relevant if integrating with other health systems.

JSON : *JavaScript Object Notation*: A lightweight data-interchange format used to transmit data between the frontend and backend.

JWT : *JSON Web Token*: A compact, URL-safe token used for securely transmitting information between parties, typically for authentication in web applications.

MERN Stack : *MongoDB, Express.js, React.js, Node.js*: The technology stack used for developing the Autism Care platform, enabling full-stack JavaScript development.

Offline-first : A design approach for PWAs where the system is functional even without a network connection, with queued actions synced when connectivity is restored.

PWA : *Progressive Web Application*: A web application that uses modern web capabilities to deliver an app-like experience, including offline access, push notifications, and installability.

PHI : *Protected Health Information*: Any personal data related to the health status, provision of healthcare, or payment for healthcare of an individual, especially sensitive for children in this system.

REST : *Representational State Transfer*: An architectural style for designing networked applications, often used in web APIs.

S3 : *Simple Storage Service*: Cloud-based object storage used for storing unstructured data like images, PDFs, and videos.

UUID : *Universally Unique Identifier*: A 128-bit identifier used to uniquely identify records in a database or objects across distributed systems.

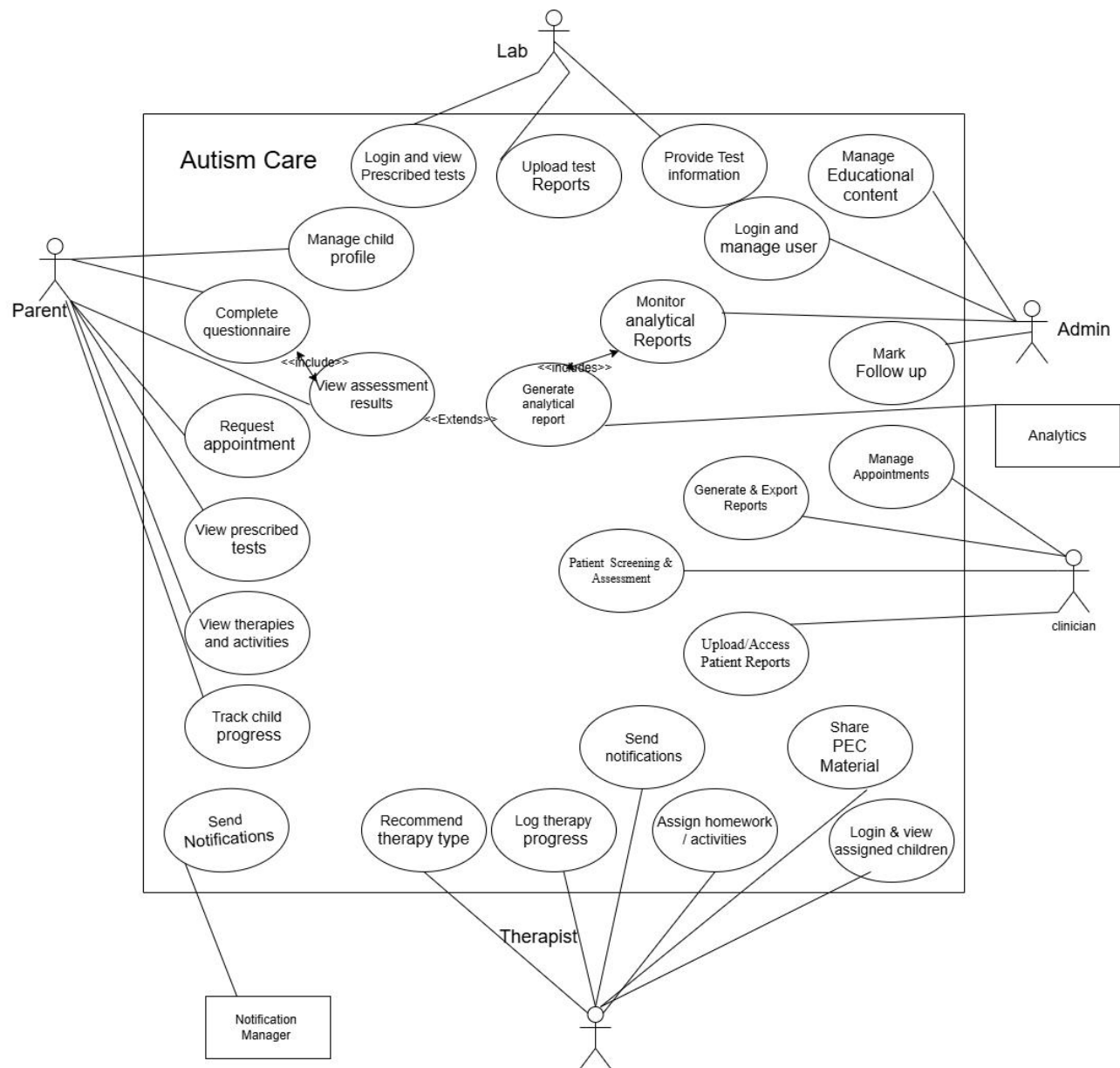
Workbox : A set of libraries for building service workers in PWAs, enabling caching, offline access, and background synchronization.

Appendix B: Analysis Models

This appendix presents the analysis models that support and clarify the system requirements for the Autism Care platform. These diagrams provide a structured visual representation of the system's functionality, data organization, and interaction flows. The included models—such as use case diagrams, class diagrams, sequence diagrams, activity diagrams, and the entity-relationship diagram—aid stakeholders in understanding how the system operates and how its components relate to each other. These models serve as a foundation for system design, development, and verification.

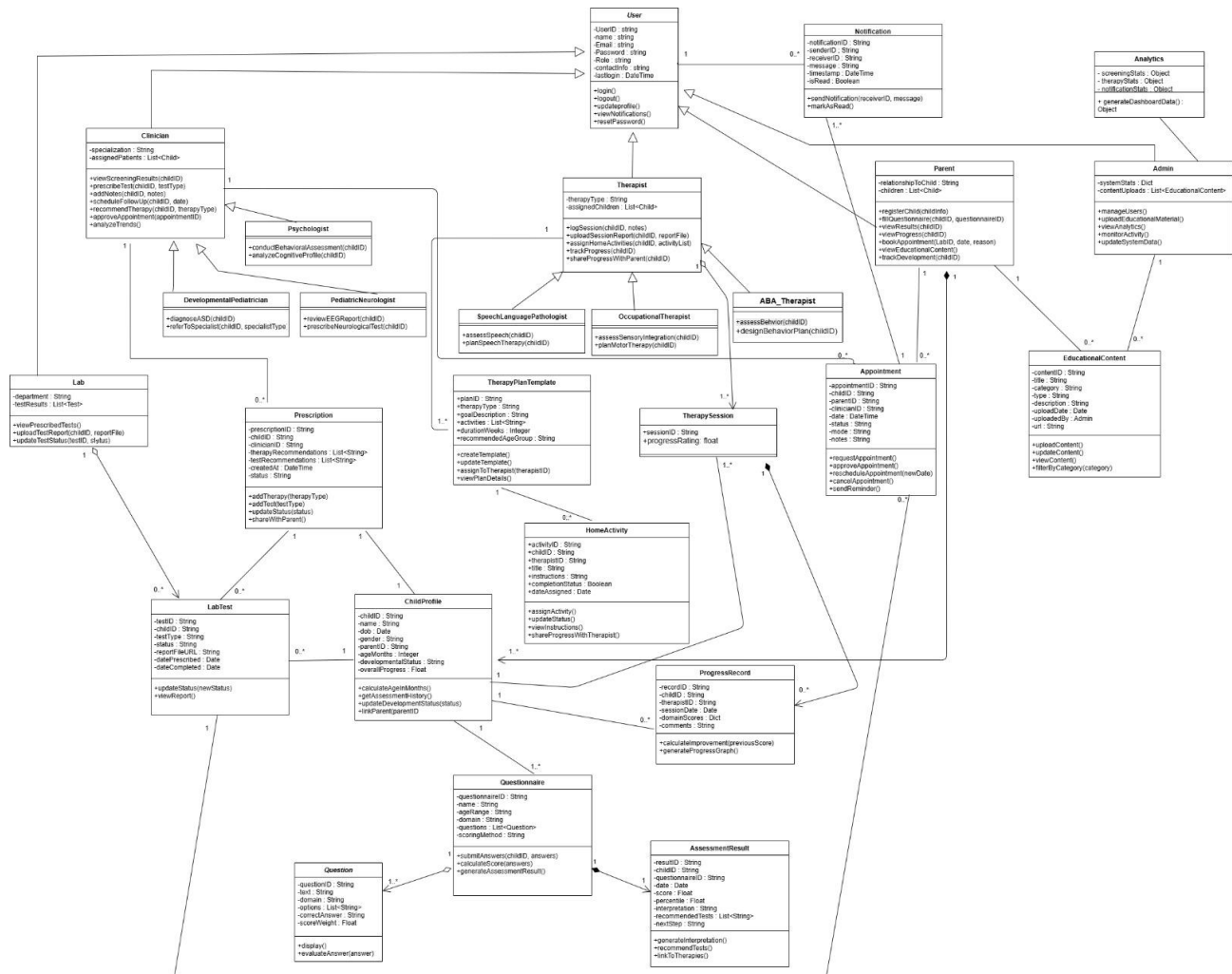
UseCase Diagram

Figure 2: Use-Case Diagram



Class Diagram

Figure 3 :Class Diagram



Activity Diagrams

Figure 4.0: Admin Approval Workflow

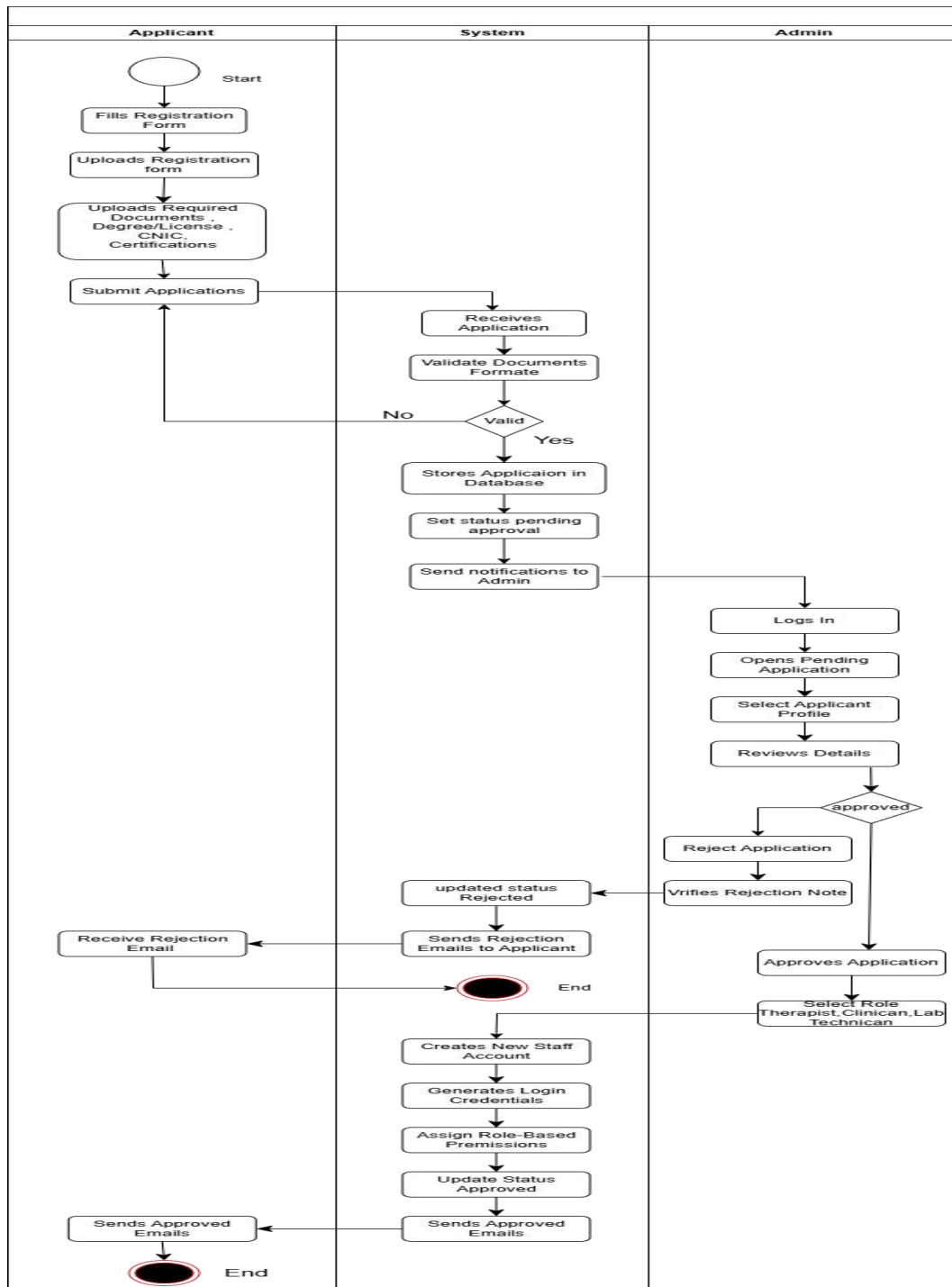


Figure 4.1: Screening workflow

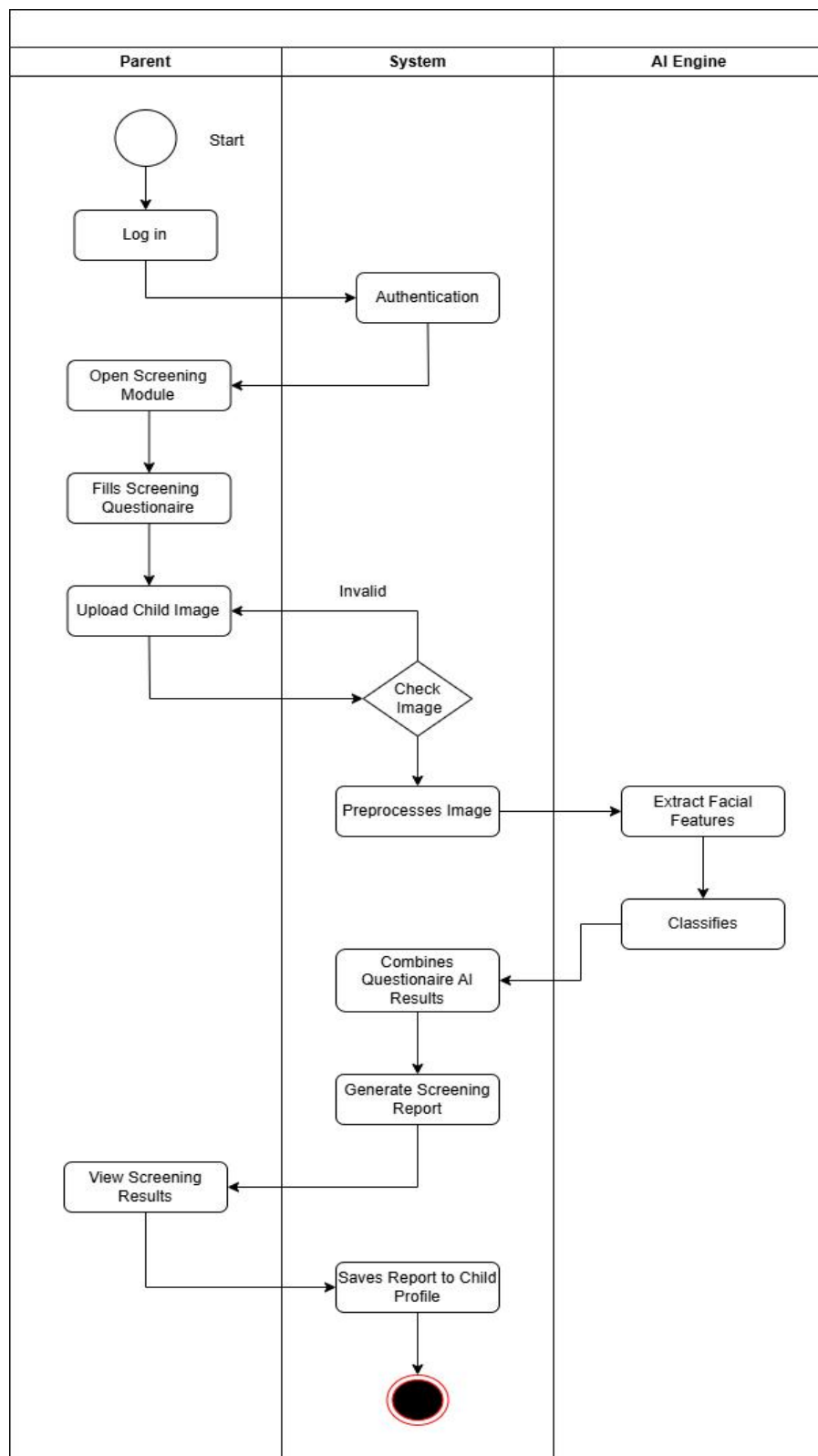


Figure 4.2: Clinician Module workflow

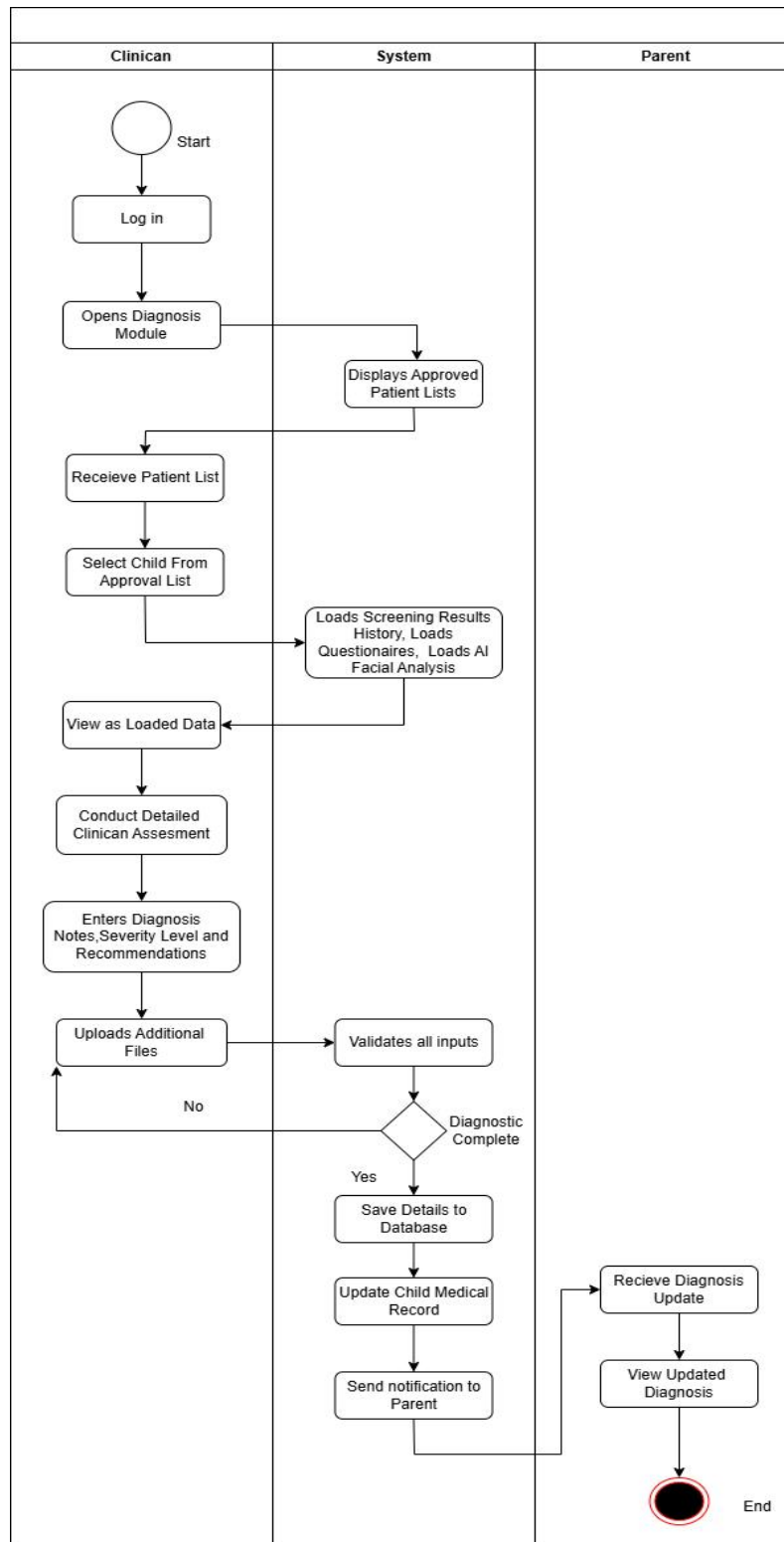


Figure 4.3: Lab workflow

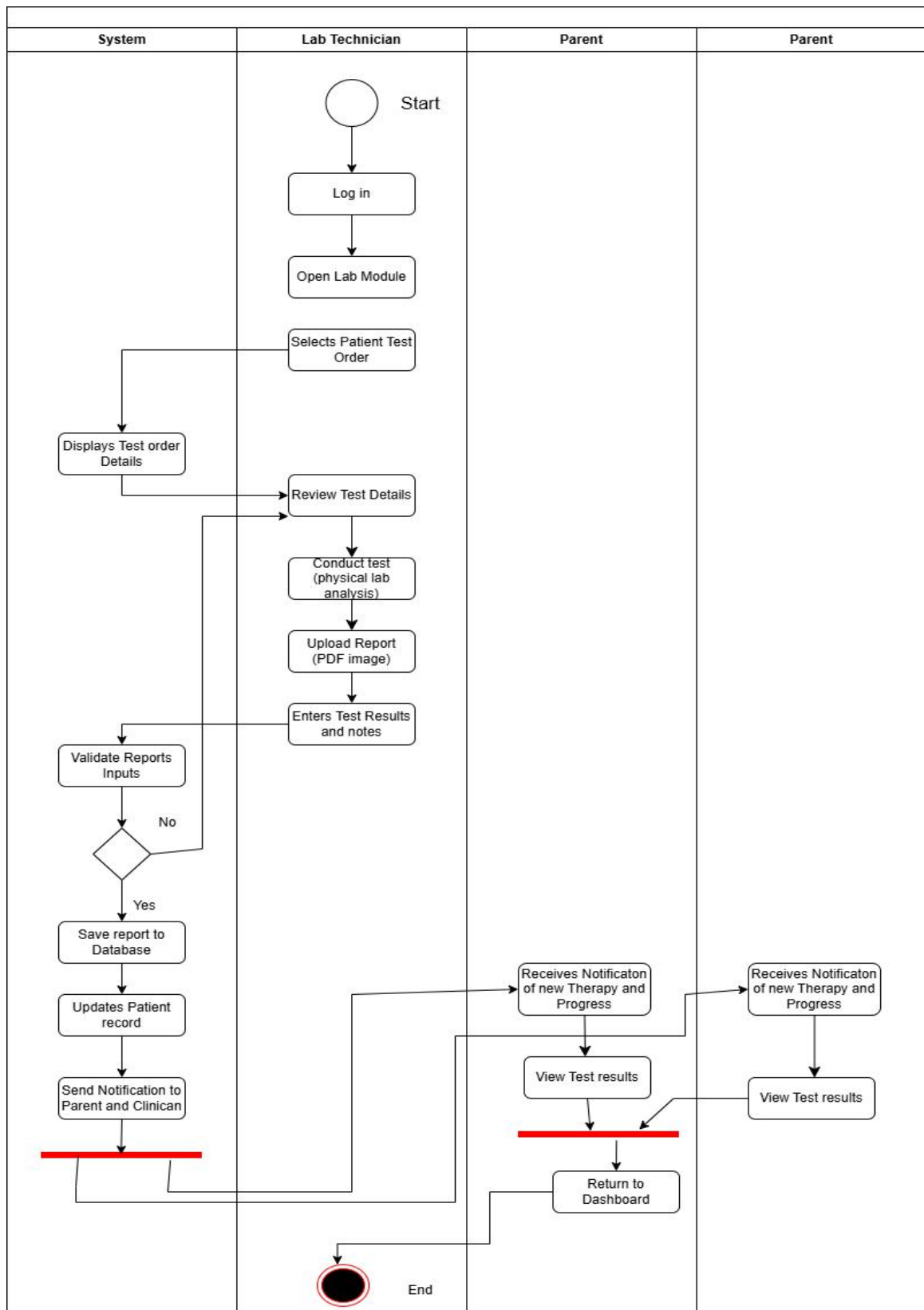


Figure 4.4: Appointment booking workflow

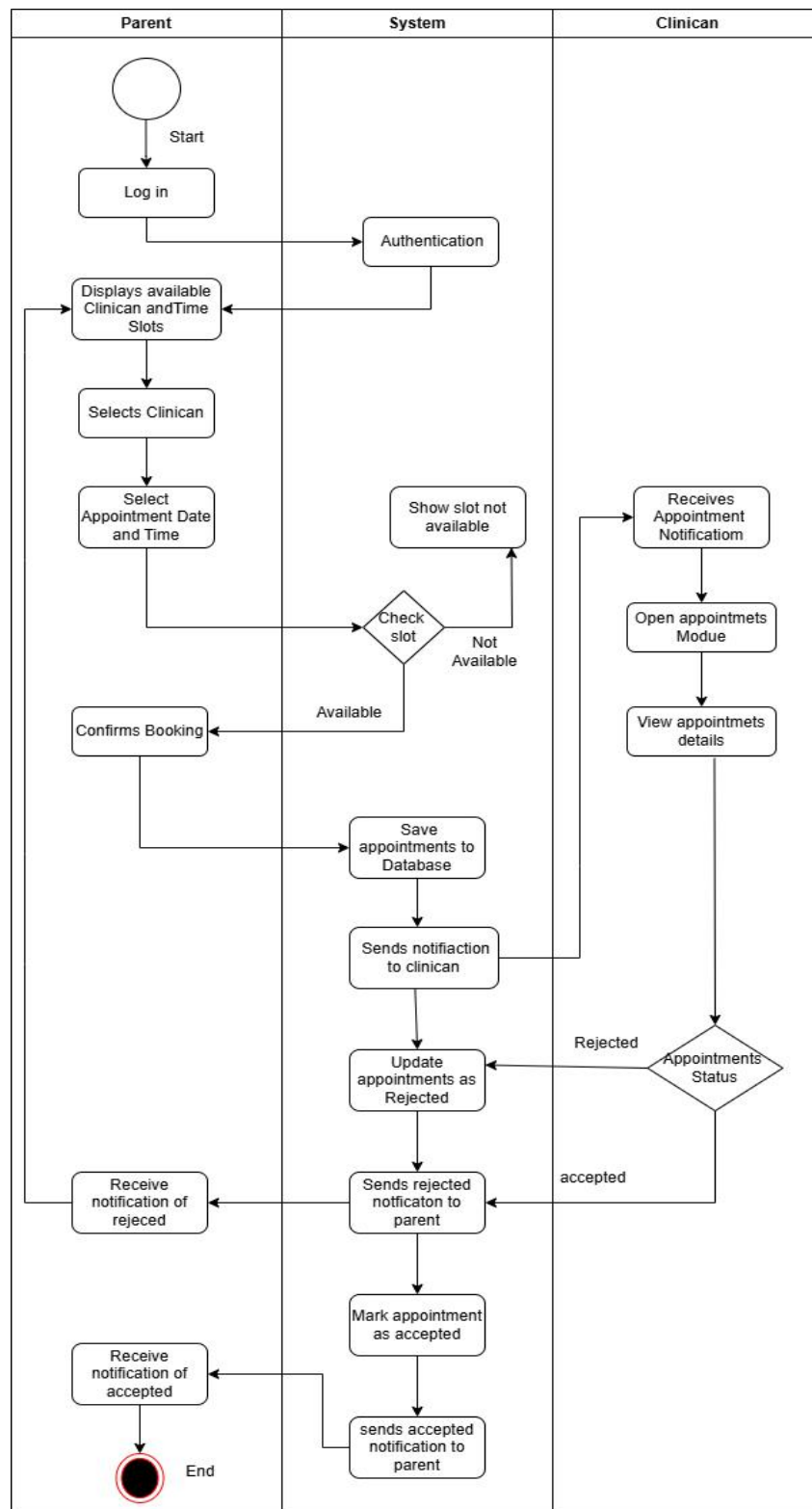
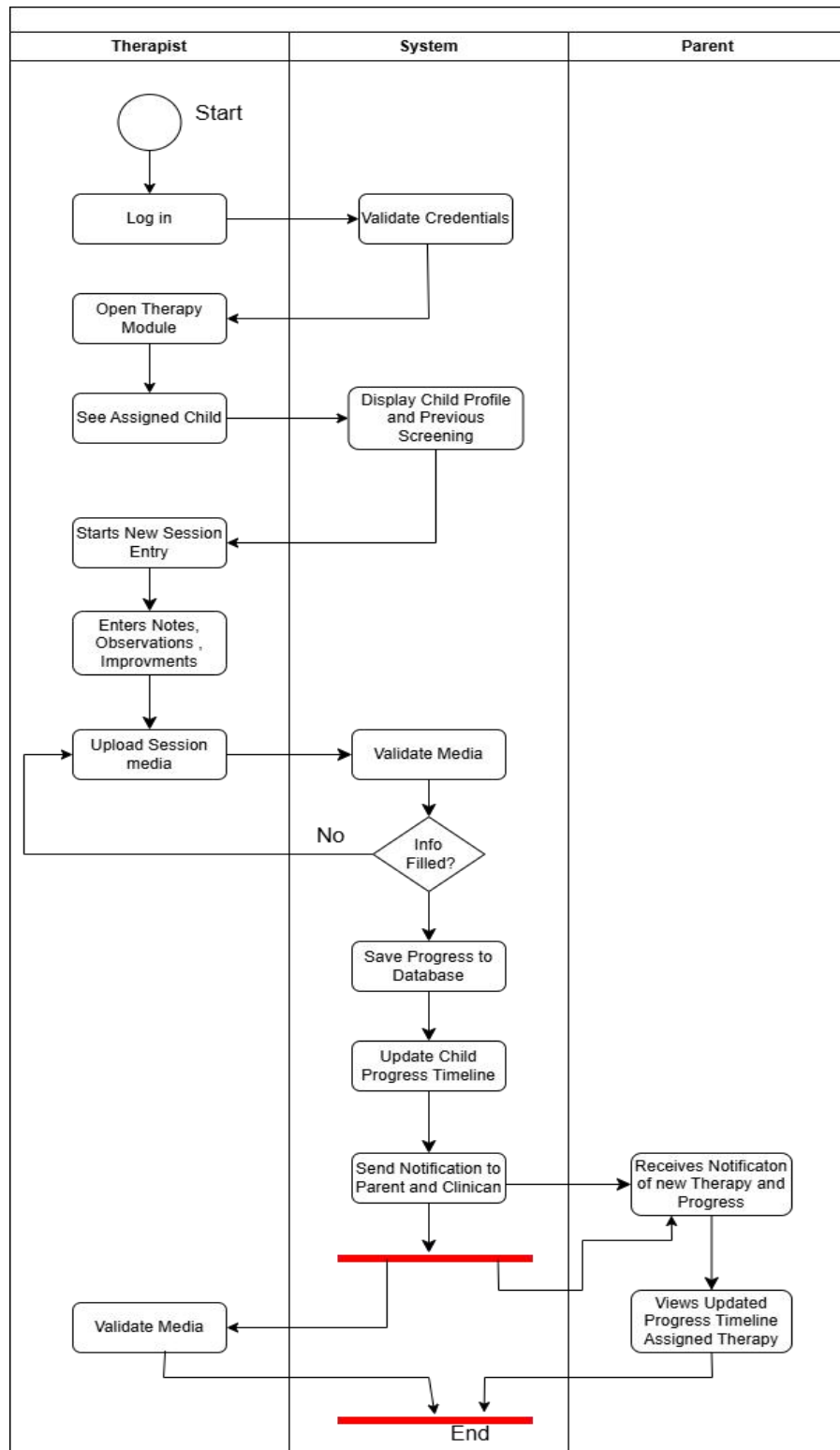


Figure 4.5: Therapy Module workflow



Sequence Diagrams

Figure 5.0: Sequence of whole system

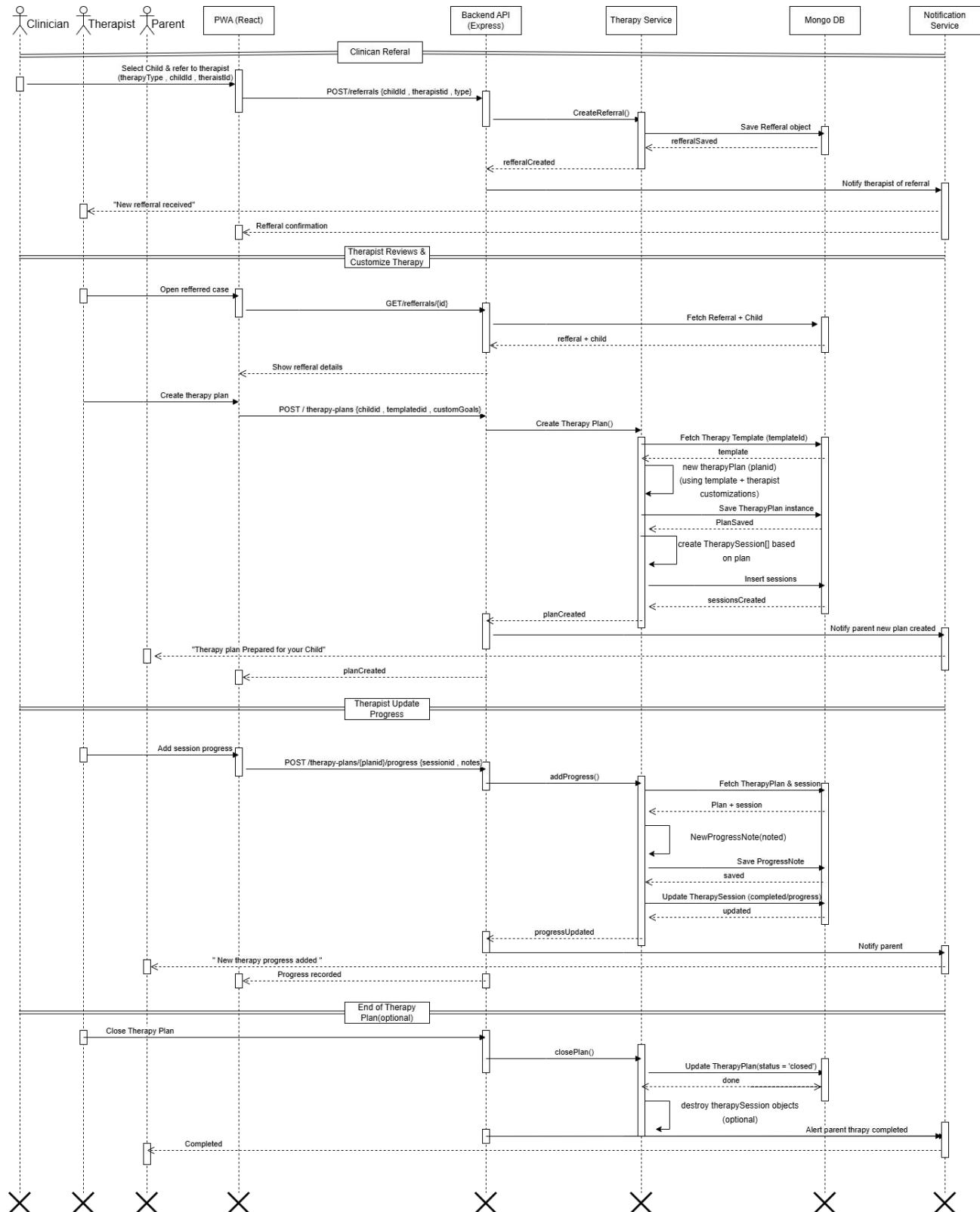


Figure 5.1: Admin Module Sequence

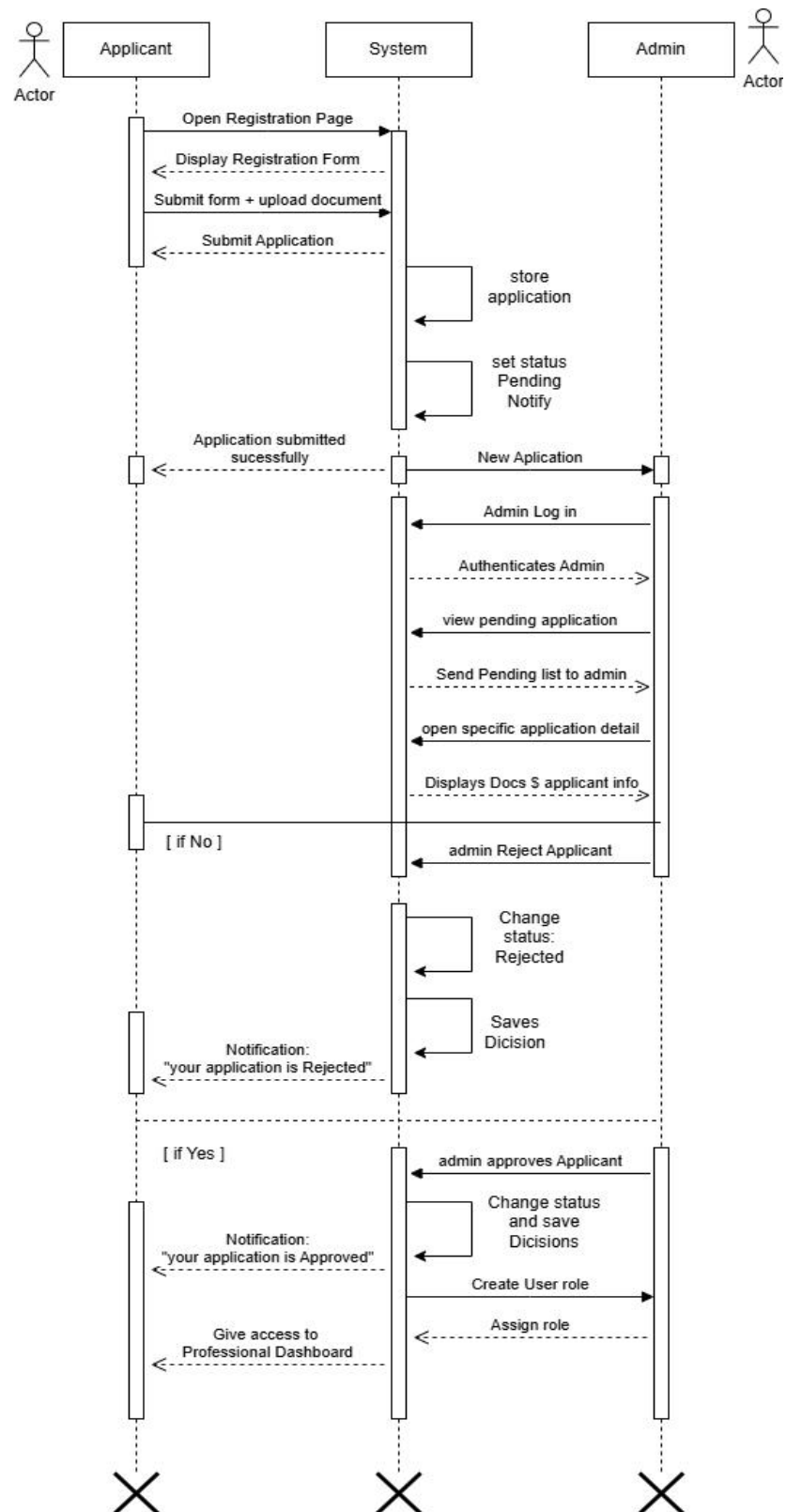


Figure 5.2: Screening Sequence

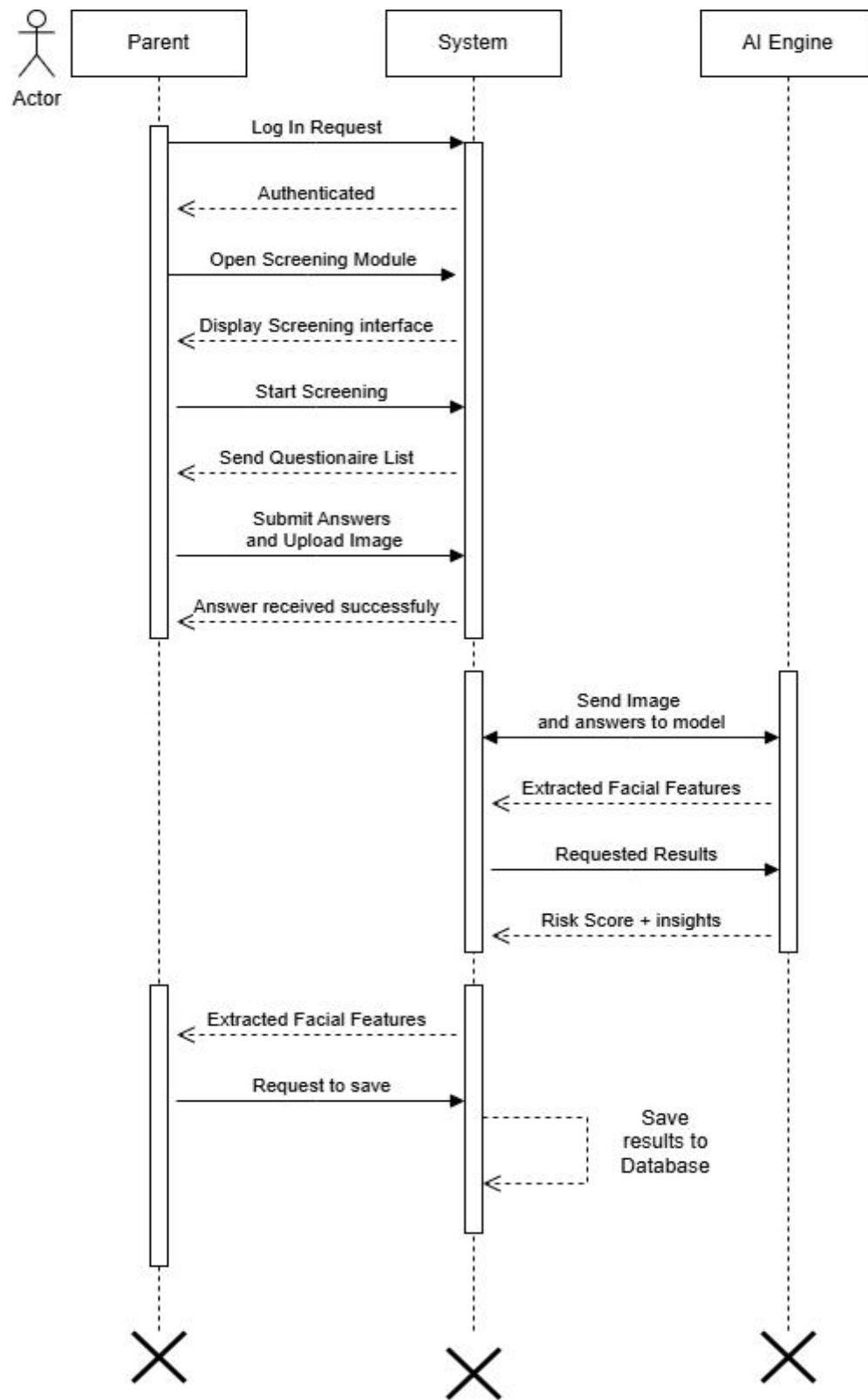
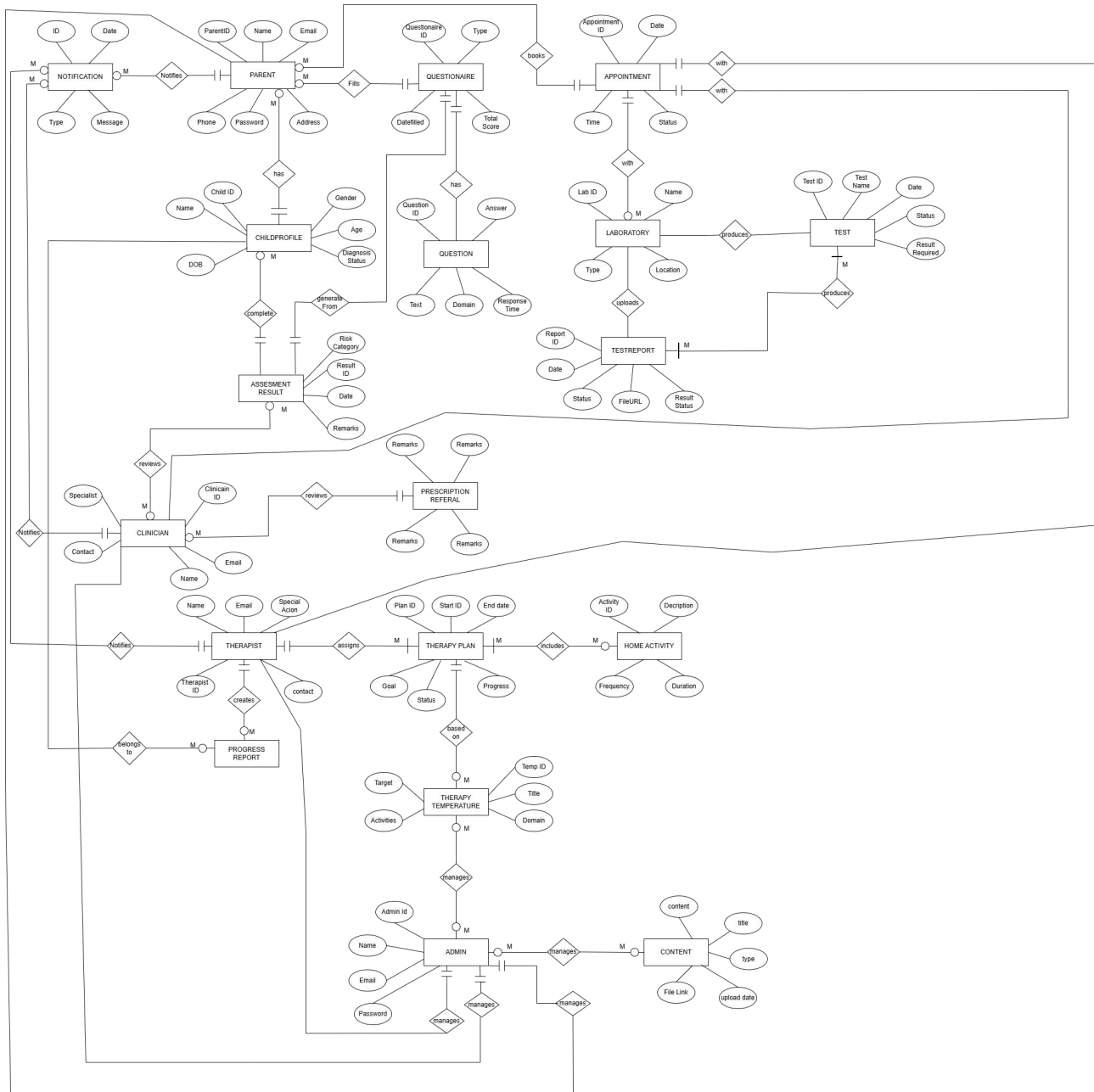
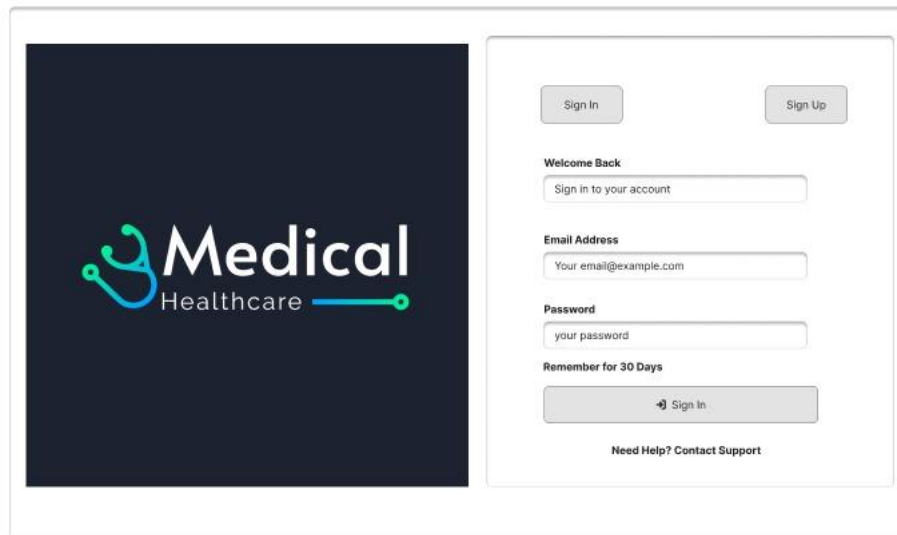


Figure 6:ER-Diagram



Dashboard Designs

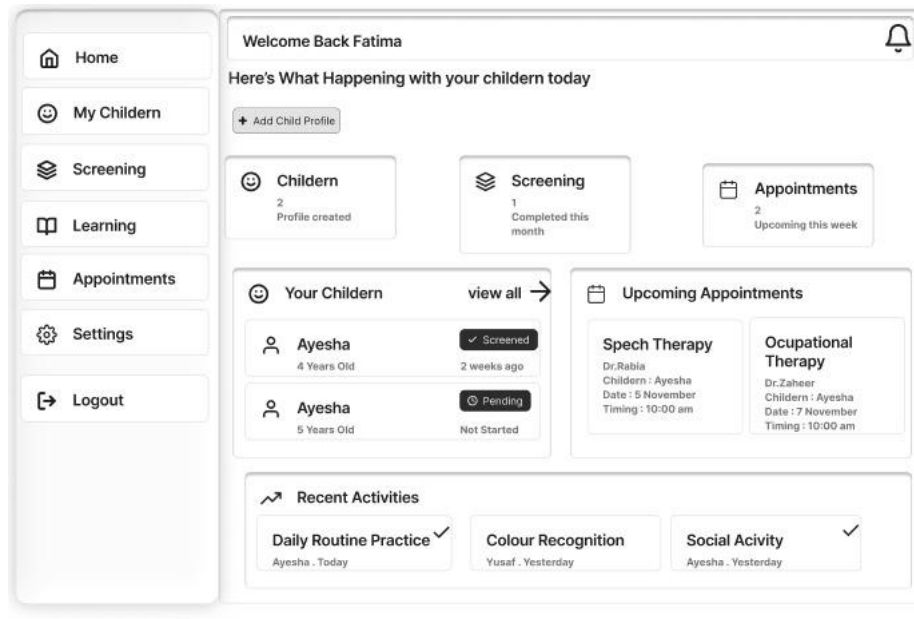
Figure 7.0:Login



The login form is titled "Medical Healthcare" and features a dark blue header with a logo. The main content area is white and contains the following elements:

- Sign In / Sign Up:** Two buttons at the top right.
- Welcome Back:** A section with a "Sign in to your account" input field.
- Email Address:** A section with a "Your email@example.com" input field.
- Password:** A section with a "your password" input field.
- Remember for 30 Days:** A checkbox and a "Sign In" button.
- Need Help? Contact Support:** A link at the bottom.

Figure 7.1:Parent Dashboard



The parent dashboard is titled "Welcome Back Fatima" and features a sidebar with navigation links: Home, My Children, Screening, Learning, Appointments, Settings, and Logout. The main content area is divided into several sections:

- Welcome Back Fatima:** A header with a bell icon.
- Here's What Happening with your children today:** A section with an "Add Child Profile" button.
- Children:** A section with 2 "Profile created" items.
- Screening:** A section with 1 "Completed this month" item.
- Appointments:** A section with 2 "Upcoming this week" items.
- Your Children:** A section with a "view all" link and two child profiles: Ayesha (4 Years Old, Screened 2 weeks ago) and Ayesha (5 Years Old, Pending Not Started).
- Upcoming Appointments:** A section with two appointments: Speech Therapy (Dr. Rabia, Children: Ayesha, Date: 5 November, Timing: 10:00 am) and Occupational Therapy (Dr. Zaheer, Children: Ayesha, Date: 7 November, Timing: 10:00 am).
- Recent Activities:** A section with three activities: Daily Routine Practice (Ayesha - Today), Colour Recognition (Yusuf - Yesterday), and Social Activity (Ayesha - Yesterday).

Figure 7.2: Clinician Dashboard

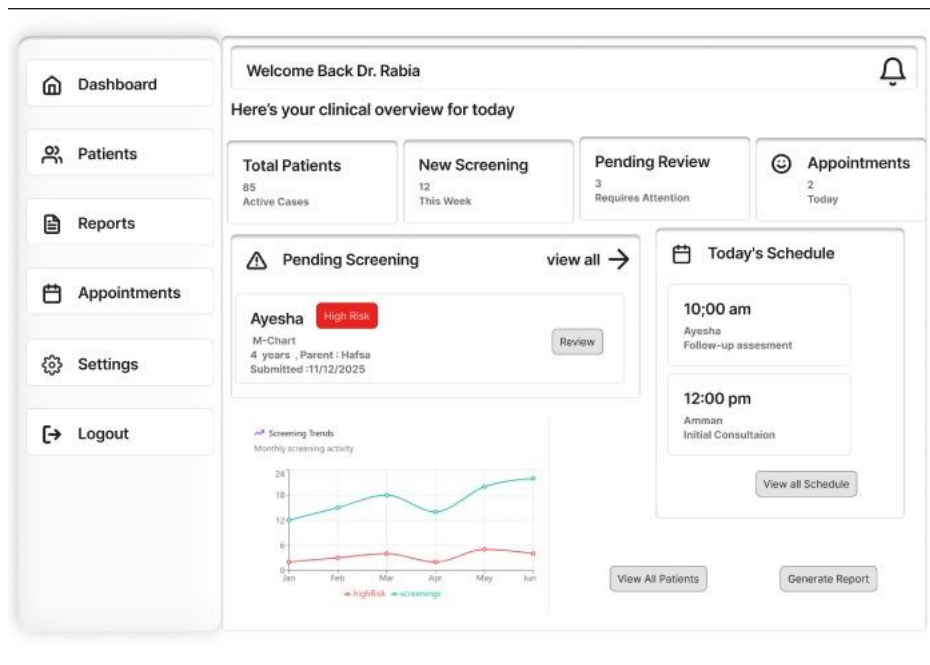


Figure 7.3:Therapist Dashboard

